Project: Hybrid Cloud Setup Using VPN or Direct Connect

Services: AWS Site-to-Site VPN, VPC, EC2, Route Tables

Goal: Connect an on-premises simulated environment (e.g., using VirtualBox) with AWS.

Skills: Routing, hybrid DNS, secure connectivity.

Definition:

The Hybrid Cloud Setup project demonstrates how to connect an on-premises network with a cloud environment to enable secure communication and resource sharing between them. In this implementation, two separate networks are created using Amazon VPC, where one VPC represents the cloud environment and the other simulates the on-premises infrastructure. Virtual machines are launched using Amazon EC2 in each network, and connectivity between them is established using Amazon VPC Peering. By configuring subnets, route tables, internet gateways, and security groups, communication between both environments is enabled and verified through network tests such as ping. This project helps in understanding hybrid cloud networking, secure connectivity, and practical implementation of AWS networking services.

Scope of the Project:

- To create a cloud network environment using Amazon VPC.
- To simulate an on-premises network using another VPC in AWS.
- To launch virtual servers using Amazon EC2 in both cloud and on-premises networks.
- To configure networking components such as subnets, route tables, and internet gateways.
- To establish connectivity between the two networks using Amazon VPC Peering.
- To configure security groups and routing rules for secure communication.
- To test network connectivity between the instances using ICMP ping commands.
- To understand the working concept of hybrid cloud architecture and AWS networking services.

Methodology:

1.Using Site-to-Site VPN

- Establish a secure connection between on-premises infrastructure and the cloud using AWS Site-to-Site VPN.
- Configure a Virtual Private Gateway and Customer Gateway in Amazon VPC.
- Connect the on-premises network router with the AWS cloud network through a VPN tunnel.
- Configure routing tables and security rules to allow network communication.
- Verify connectivity between servers running on Amazon EC2 in both environments.

2.Using VPC Peering

- Create two separate VPC networks to represent cloud and on-premises environments using Amazon VPC.
- Launch virtual servers in both networks using Amazon EC2.
- Create a connection between the two VPCs using Amazon VPC Peering.
- Update route tables to allow traffic between the VPC networks.
- Configure security groups and test connectivity using ping between the instances.

AWS Service used in the Project:

1.Amazon VPC

- **Definition:** A Virtual Private Cloud is a logically isolated virtual network in AWS where you can launch cloud resources.
- **Role:** Provides separate network environments for the cloud and simulated on-premises infrastructure.
- **Purpose in the Project:** Used to create two isolated networks (Cloud VPC and On-Prem VPC) that form the base of the hybrid cloud architecture.

2. Amazon EC2

- **Definition:** Amazon EC2 is a service that provides scalable virtual servers in the cloud.
- **Role:** Hosts the virtual machines used to simulate servers in both cloud and on-premises networks.
- **Purpose in the Project:** Used to deploy instances in each VPC to test network connectivity between the two environments.

3. Amazon VPC Peering

- **Definition:** A networking connection that allows two VPCs to communicate privately using their private IP addresses.
- **Role:** Establishes direct connectivity between the Cloud VPC and the On-Prem VPC.
- **Purpose in the Project:** Enables communication between EC2 instances in different VPC networks.

4. AWS Internet Gateway

- **Definition:** An Internet Gateway is a component that allows communication between a VPC and the internet.
- **Role:** Provides internet access to EC2 instances inside the VPC.
- **Purpose in the Project:** Allows SSH access to EC2 instances and supports external connectivity for management.

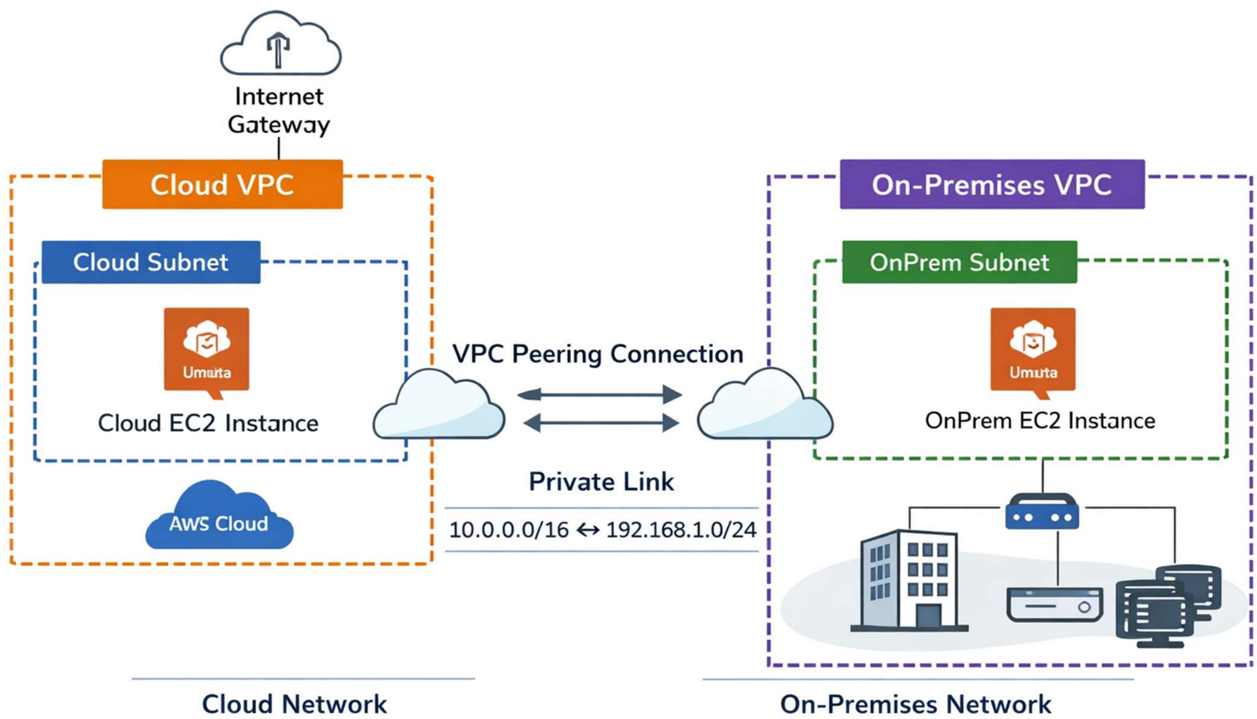
5. AWS Route Tables

- **Definition:** Route tables are a set of rules that determine how network traffic is directed within a VPC.
- **Role:** Controls how data flows between the Cloud VPC and On-Prem VPC.
- **Purpose in the Project:** Used to configure routing paths that enable communication between both networks.

6. AWS Security Groups

- **Definition:** Security groups act as virtual firewalls that control inbound and outbound traffic for EC2 instances.
- **Role:** Protects instances by allowing only specific network traffic such as SSH and ICMP.
- **Purpose in the Project:** Ensures secure communication between instances while allowing required traffic for connectivity testing.

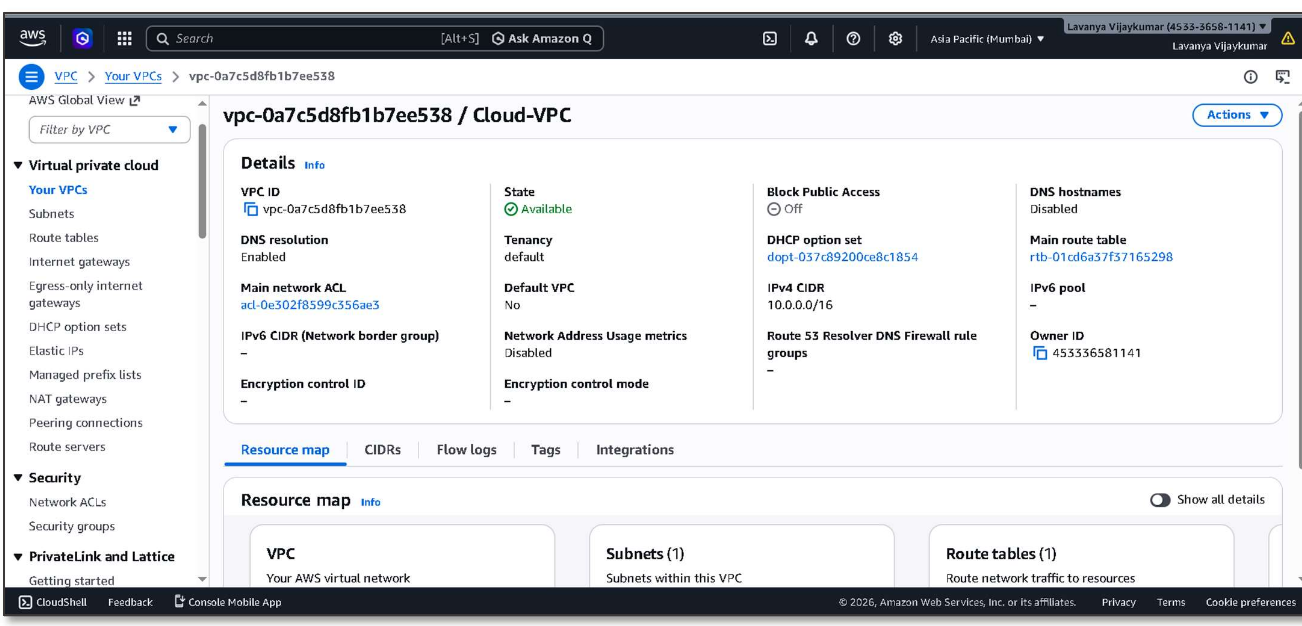
Architecture Diagram:



Project Implementation Steps:

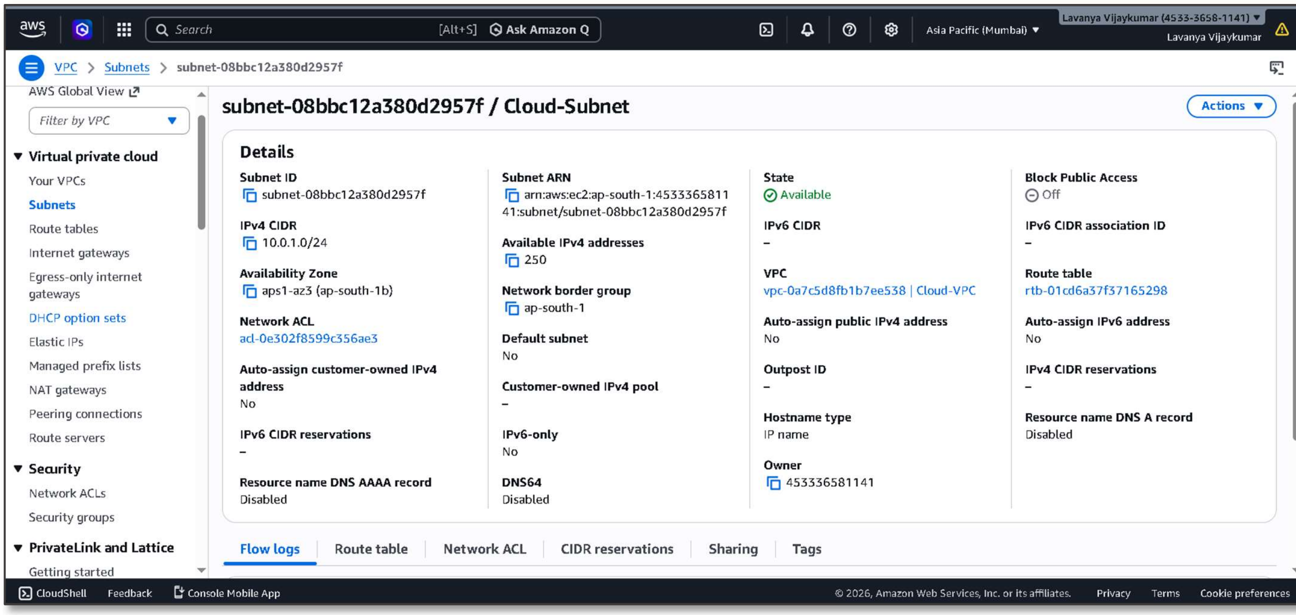
STEP 1: Create Cloud VPC

- Go to VPC Dashboard.
- Click Create VPC.
- Configuration:
- Name: Cloud-VPC
  IPv4 CIDR block: 10.0.0.0/16



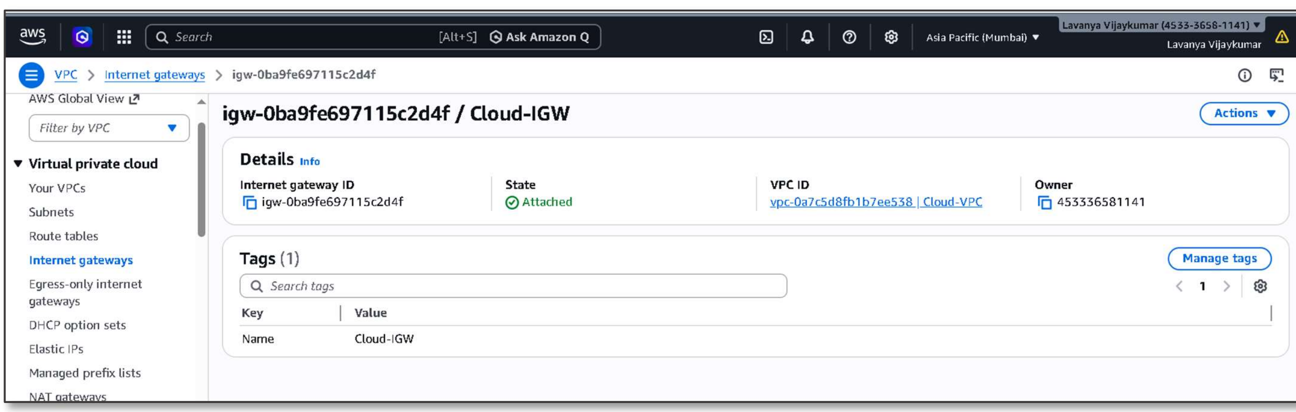
STEP 2: Create Cloud Subnet

- Create a subnet inside the Cloud VPC.
- Name: Cloud-Subnet
  VPC: Cloud-VPC
  CIDR block: 10.0.1.0/24
  Availability Zone: Any



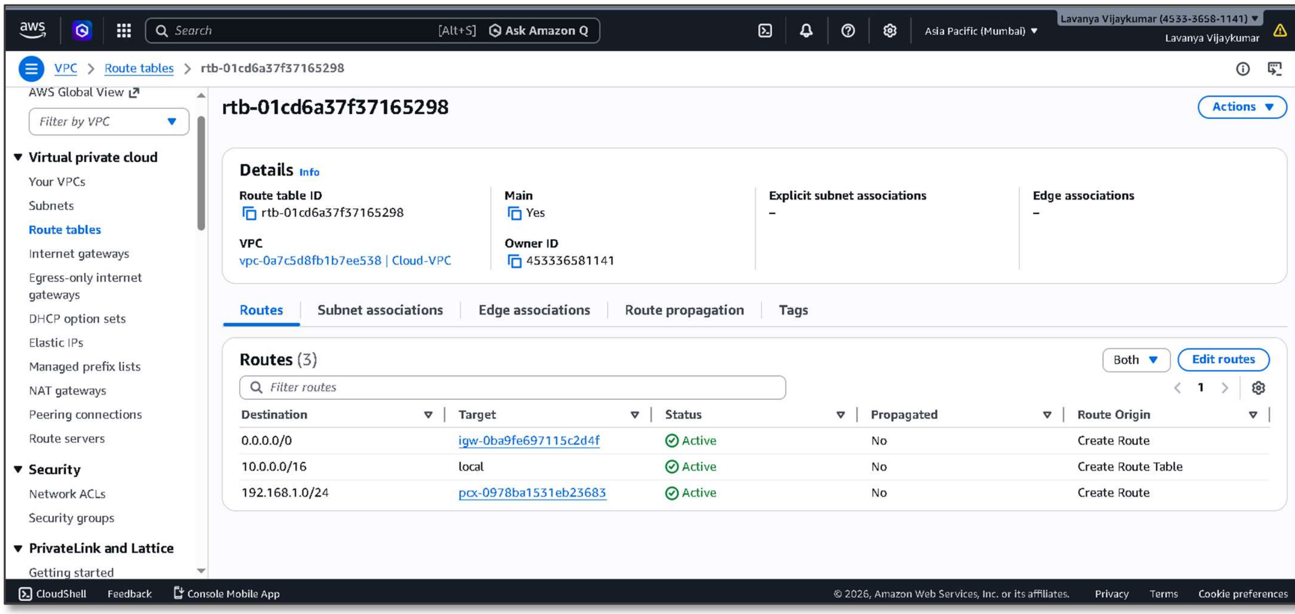
STEP 3: Create Internet Gateway for Cloud VPC

- Go to Internet Gateways.
- Click Create Internet Gateway.
  Name: Cloud-IGW
- Attach it to Cloud-VPC.



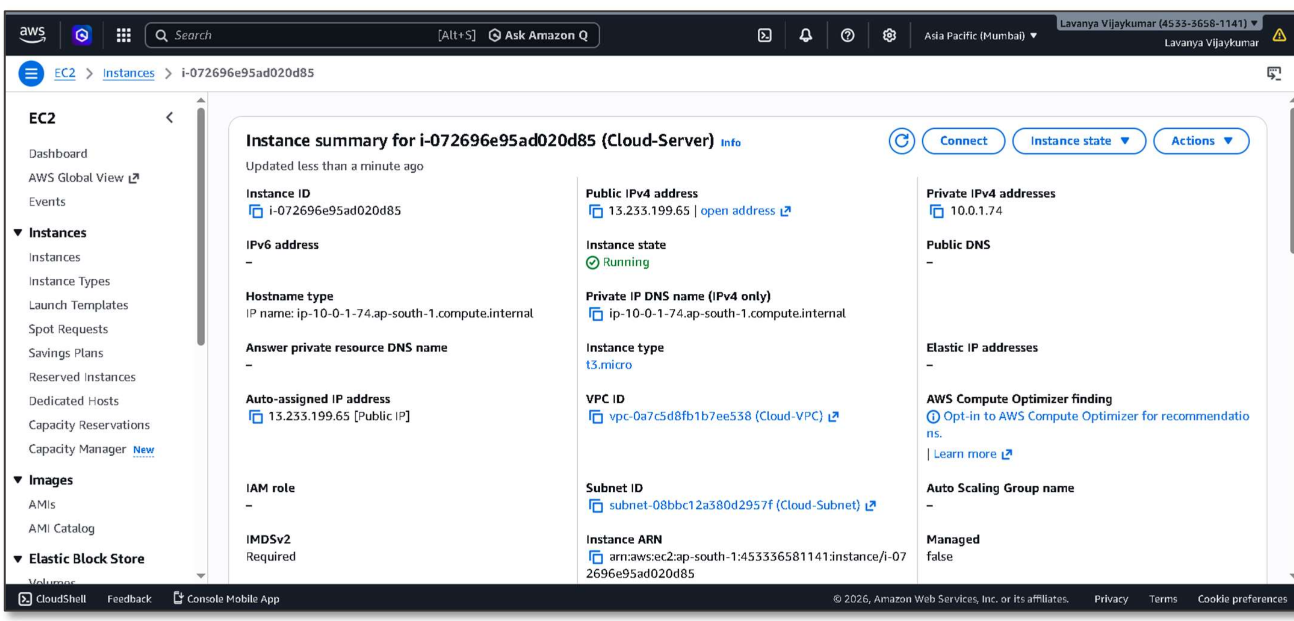
STEP 4: Configure Cloud Route Table

- Open Route Tables → Cloud-VPC Route Table → Edit Routes
- Add: Destination: 0.0.0.0/0
  Target: Internet Gateway



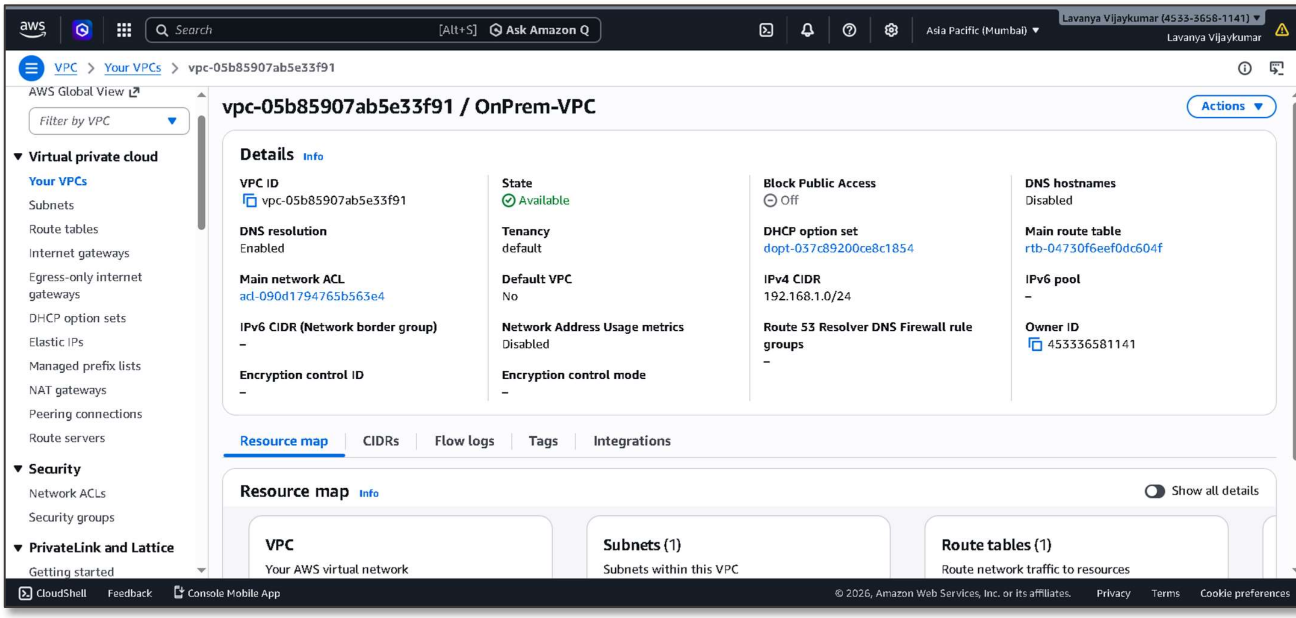
STEP:5: Launch Cloud EC2 Instance

- Go to EC2 → Launch Instance.
- Configuration:
- Instance Name: Cloud-Server
  AMI: Ubuntu
  Network: Cloud-VPC
  Subnet: Cloud-Subnet
  Auto Assign Public IP: Enabled
- Allow security group rules:
  SSH (22)
  ICMP



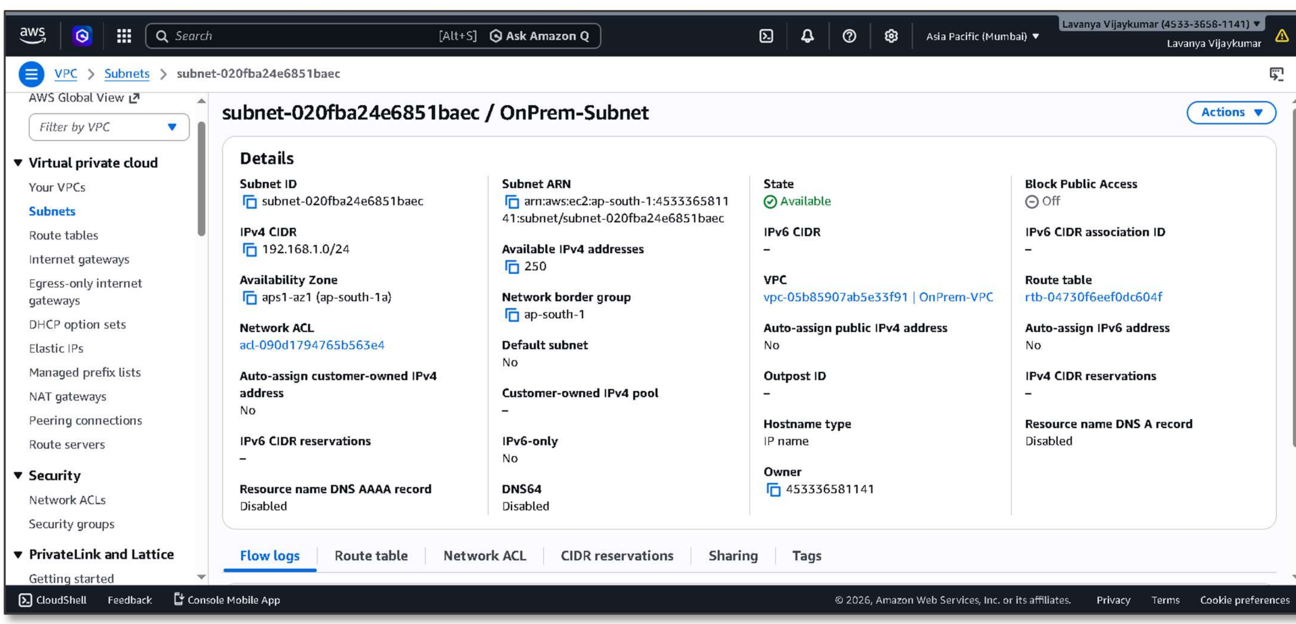
STEP 6: Create On-Premises VPC

- Create another VPC to simulate the on-premises network.
- Name: OnPrem-VPC
  CIDR block: 192.168.1.0/24



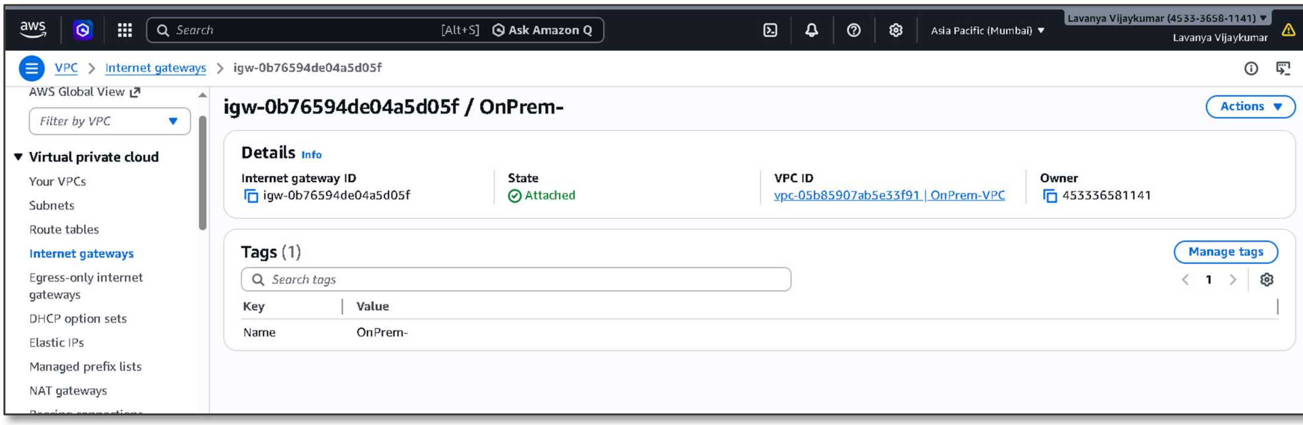
STEP 7: Create On-Premises Subnet

- Name: OnPrem-Subnet
  VPC: OnPrem-VPC
  CIDR block: 192.168.1.0/24



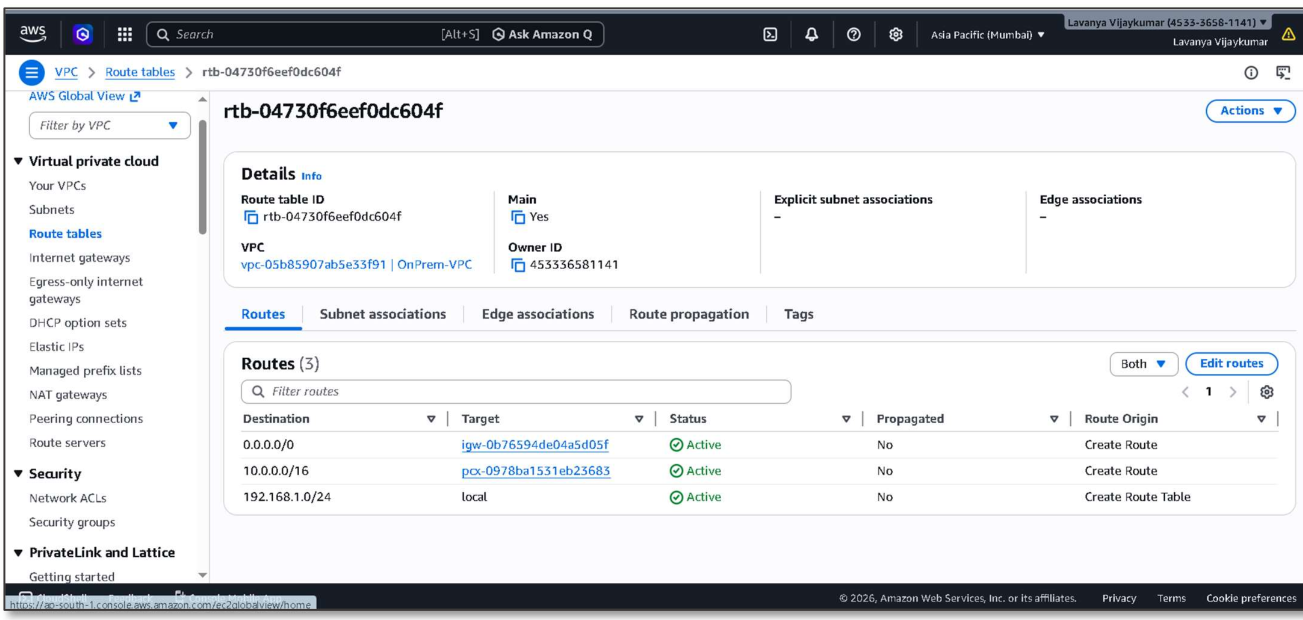
STEP 8: Create Internet Gateway for On-Prem VPC

- Create and attach:
  Name: OnPrem-IGW
  Attach to: OnPrem-VPC



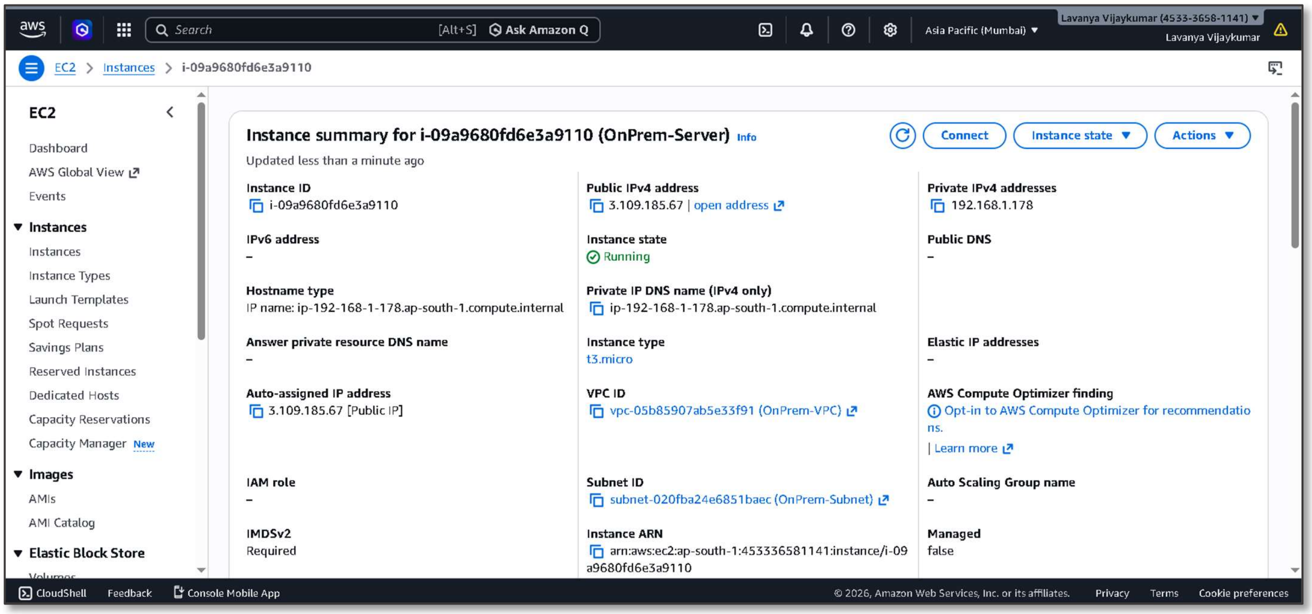
STEP 9: Configure On-Prem Route Table

- Add route:
  Destination:0.0.0.0/0
  Target: Internet Gateway



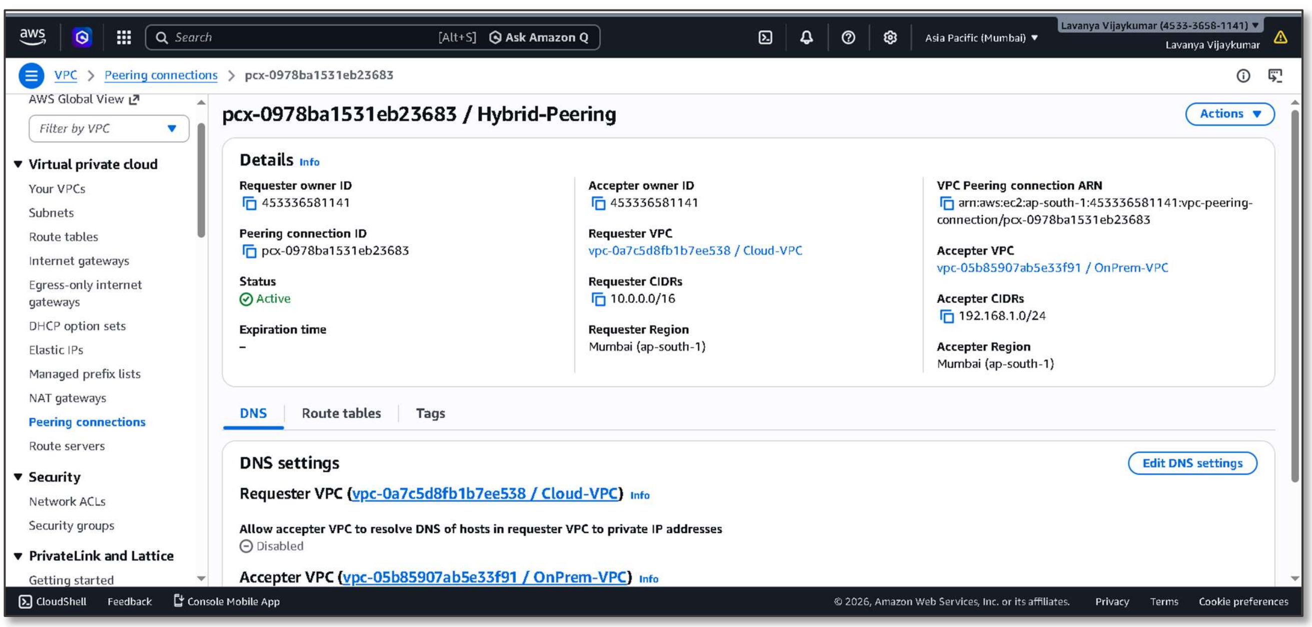
STEP 10: Launch On-Prem EC2 Instance

- Configuration:
- Instance Name: OnPrem-Server
  AMI: Ubuntu
  Network: OnPrem-VPC
  Subnet: OnPrem-Subnet
  Auto Assign Public IP: Enabled
- Allow security rules:
  SSH
  ICMP



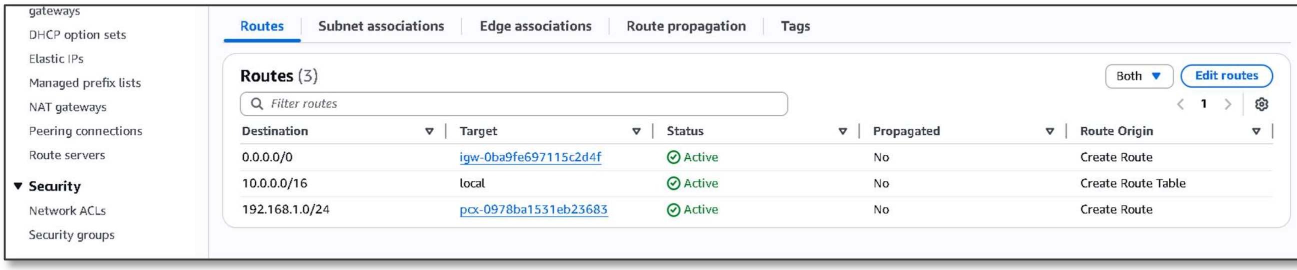
STEP 11: Create VPC Peering Connection

- To connect both networks:
- Create **VPC Peering**.
- Name: Hybrid-Peering
  Requester VPC: Cloud-VPC
  Accepter VPC: OnPrem-VPC
- Accept the request.



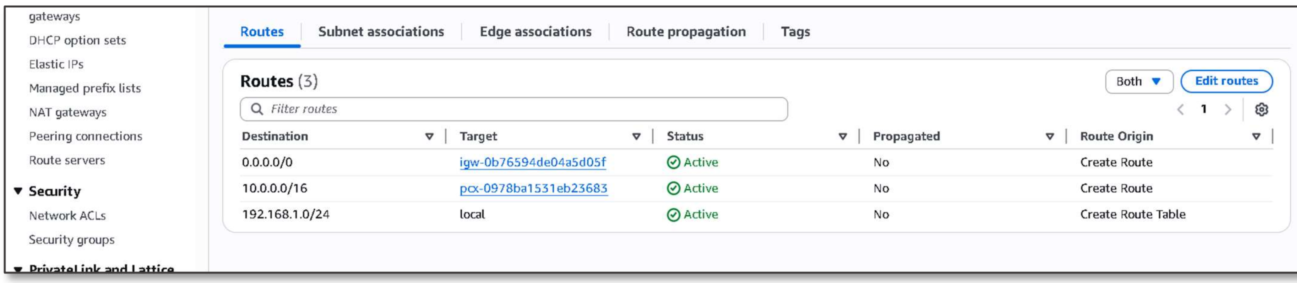
Step 12: Update Cloud Route Table

- Add route:
- Destination:192.168.1.0/24
  Target: VPC Peering Connection



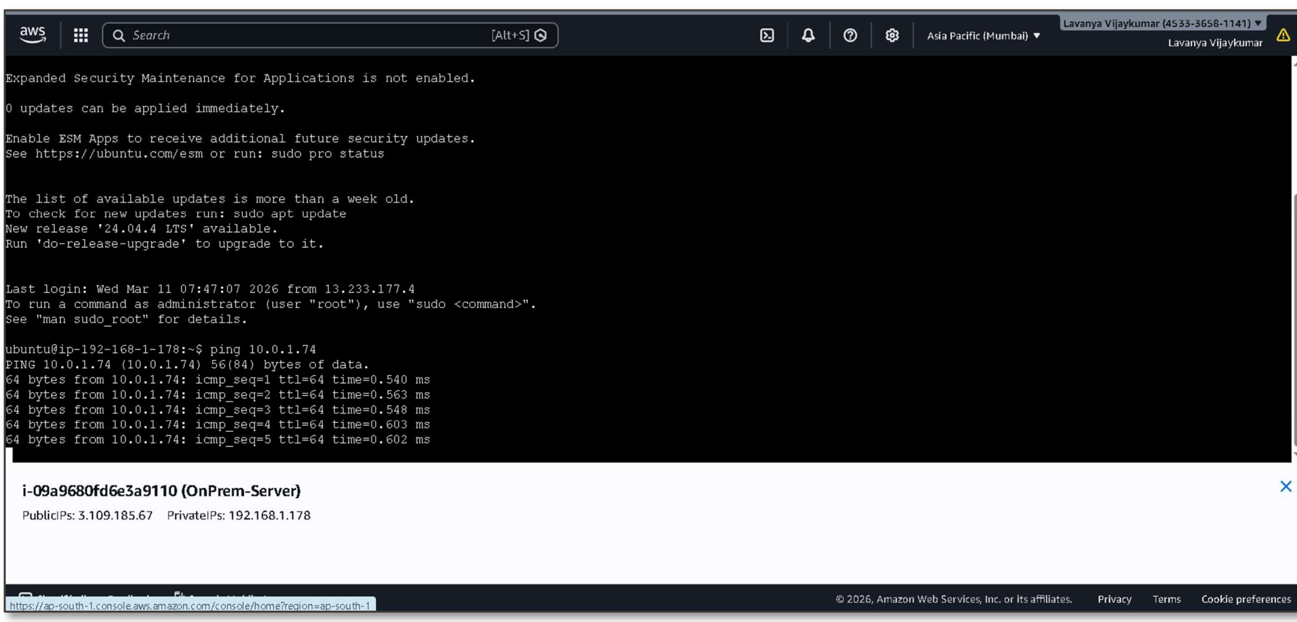
STEP 13: Update On-Prem Route Table

- Add route:
- Destination: 10.0.0.0/16
  Target: VPC Peering Connection

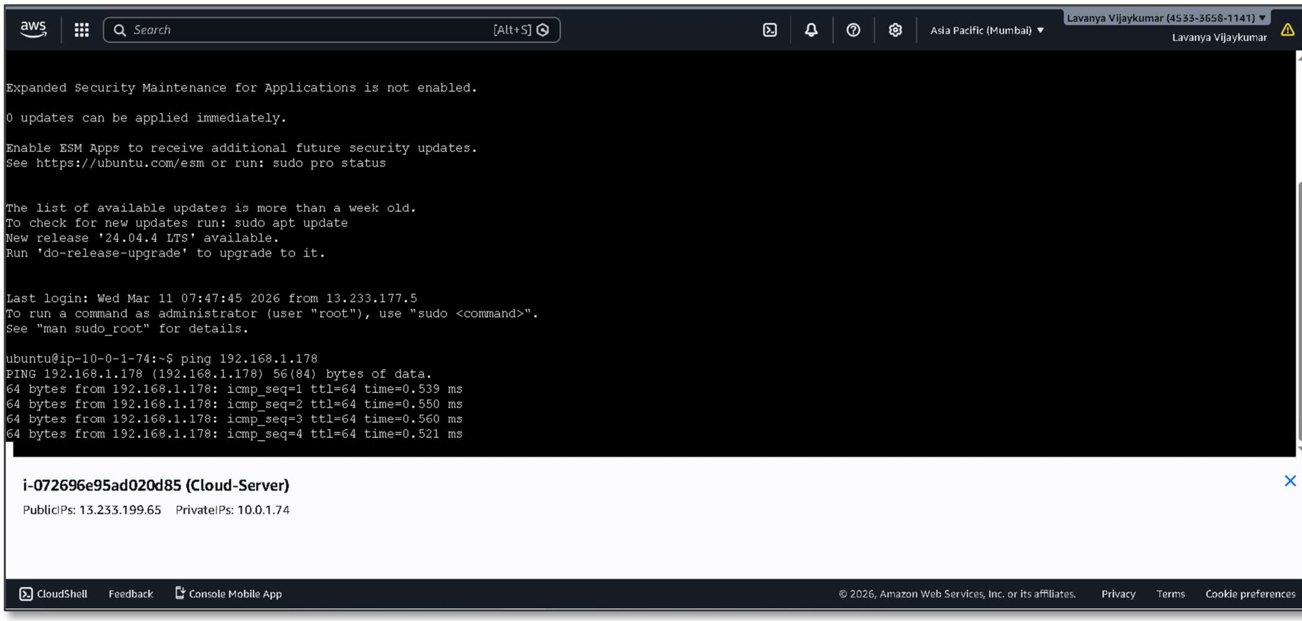


STEP 14: Test Hybrid Connectivity

- Login to the On-Prem EC2 instance
- Ping Cloud ec2 with a private IP address



- Then login to Cloud EC2 instance
- Ping On Prem ec2 with a private IP address



Troubleshooting:

1.Issue: Unable to access the internet from the instance

- **Cause:** Internet Gateway is not attached to the VPC or the route table does not have an internet route.
- **Solution:** Attach an AWS Internet Gateway to the VPC and add a route 0.0.0.0/0 in the route table pointing to the Internet Gateway.

Conclusion:

In this project, a hybrid cloud networking environment was successfully designed and implemented using Amazon VPC and Amazon EC2. Two separate networks were created to represent the cloud and the on-premises environments, and connectivity between them was established using Amazon VPC Peering. Proper configuration of subnets, route tables, internet gateways, and security groups enabled secure communication between the instances in both networks. The connectivity was successfully verified through ping tests, demonstrating effective network communication. This project helped in understanding hybrid cloud architecture, AWS networking components, and the practical implementation of secure connectivity between different network environments.